

Two-Stream Instability Observatory: An Interactive Theory Laboratory and Live NASA MMS Data Monitor

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Software

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Summary

The two-stream instability (TSI) is a fundamental collective phenomenon in plasma physics, occurring when two counter-propagating electron beams transfer energy to electrostatic waves, leading to exponential growth of Langmuir wave amplitudes. Understanding TSI is critical across astrophysics, magnetospheric physics, laser-plasma interactions, and inertial confinement fusion research (Buneman, 1958; Weibel, 1959).

The **Two-Stream Instability Observatory** is an open-source, browser-based research platform consisting of two tightly integrated components: a **Theory Laboratory** providing interactive exploration of exact analytical dispersion relations validated against SMILEI particle-in-cell (PIC) simulations (Derouillat et al., 2018), and a **Live MMS Observatory** that downloads real NASA Magnetospheric Multiscale (MMS-1) spacecraft data every six hours and tests whether theoretical TSI conditions are active in the Earth's magnetosphere (Burch et al., 2016).

Statement of Need

Teaching and researching plasma instabilities presents a persistent challenge: the gap between analytical theory and observable data is wide, the mathematics is non-trivial, and computational tools (PIC codes, data reduction pipelines) require significant setup. No existing tool bridges all three simultaneously — interactive dispersion theory, PIC simulation comparison, and live spacecraft data validation — in a single zero-install environment.

Existing resources each address only part of the problem. Standalone PIC codes such as SMILEI (Derouillat et al., 2018) require a local HPC environment and Python post-processing expertise. Browser-based particle simulations (e.g., particleincell.com) show raw particle dynamics but provide no analytical growth rate predictions or spacecraft data. NASA's CDAWeb interface provides MMS data but requires custom analysis pipelines. This tool closes all three gaps simultaneously for students, educators, and researchers alike.

Theory Laboratory

The laboratory implements four closed-form dispersion relations derived from the two-fluid equations for counter-propagating symmetric electron beams (Shrivastav et al., 2025, in preparation):

37 Equation 18 — Non-relativistic single-species (ions frozen):

$$\gamma = \sqrt{\frac{1}{2} \left[2v_0^2 k^2 + \omega_p^2 - \sqrt{8v_0^2 k^2 \omega_p^2 + \omega_p^4} \right]}$$

38 Equation 19 — Relativistic single-species:

$$\gamma = \sqrt{\frac{1}{2} \left[2v_0^2 k^2 + \frac{\omega_p^2}{\gamma_0^3} - \sqrt{\frac{8v_0^2 k^2 \omega_p^2}{\gamma_0^3} + \frac{\omega_p^4}{\gamma_0^6}} \right]}$$

39 Equations 20–21 extend the above to mobile-ion (multi-species) cases, replacing ω_p^2 with
40 $2\omega_p^2$ in the dispersion bracket. Here $\gamma_0 = 1/\sqrt{1-v_0^2}$ is the Lorentz factor and normalised
41 units ($c = \omega_{pe} = 1$) are used throughout.

42 All four equations are implemented as real-time JavaScript computations with interactive
43 sliders for beam velocity v_0 , plasma frequency ω_p , and wavenumber k . Growth rate
44 curves $\gamma(k)$ are rendered via Plotly (Plotly Technologies Inc., 2015). Equations 18–21
45 are validated against embedded SMILEI PIC simulation data for $v_0 = 0.1c$ to $0.95c$,
46 showing agreement within 5% across all regimes. The laboratory includes a Simulation
47 Box Designer that computes SMILEI-compatible parameters (box length, cell resolution,
48 CFL condition, saturation time estimate) from user inputs.

49 MMS Live Observatory

50 A Python pipeline (fetcher.py, 814 lines) runs automatically every six hours via GitHub
51 Actions, downloading MMS-1 Level 2 data from the NASA CDAS HAPI API (Annex &
52 others, 2024). Four instruments are queried:

Instrument	Dataset	Quantity	Cadence
FGM	MMS1_FGM_SRVY_L2	Magnetic field B (GSE)	8 sps
EDP	MMS1_EDP_FAST_L2_DCE	Electric field E (GSE)	32 sps
FPI-DES	MMS1_FPI_FAST_L2_DES- n_e , V_e , $T_{\parallel}$, $T_{\perp}$ MOMS		~4.5 s
FPI-DIS	MMS1_FPI_FAST_L2_DIS- n_i , T_i MOMS		~4.5 s

53 From these measurements, the pipeline computes: (1) theoretical TSI growth rates using
54 Equations 18–21 with the measured $v_0 = V_e/c$ and $\omega_{pe} = \sqrt{n_e e^2 / m_e \epsilon_0}$; (2) electron hole
55 detection via bipolar $E_{\parallel}$ pulse identification (Graham et al., 2016; Steinvall et al., 2021);
56 (3) bump-on-tail proxy $v_{\text{beam}}/v_{\text{thermal}}$ (Jebaraj & others, 2025); (4) TSI→Weibel causal
57 chain test via cross-correlation of $V_e(t)$ and $T_{\perp}/T_{\parallel}(t)$ (Innocenti et al., 2017); (5) Langmuir
58 wave proxy from E-field PSD near f_{pe} ; and (6) magnetic turbulence spectral index α from
59 FGM power spectra (Kolmogorov, 1941).

60 Results are stored as rolling JSON files (up to 720 records, approximately six months of
61 history) committed automatically to the repository, and rendered live by the dashboard at
62 vivek-shrivastav.github.io/Two_Stream_Lab.

Educational Design

The tool serves both research and pedagogy. Section 01 includes a Beginner/Expert mode toggle, presenting the same physics at two levels of mathematical depth. FAQ entries with expandable answers cover nine conceptual questions spanning instability onset, the role of relativity, multi-species dynamics, SMILEI parameter mapping, and the distinction between TSI and Weibel instabilities. A Parameter Playground allows users to save, compare, and overlay multiple simulation runs with CSV export. Phase-space animation visualises particle trapping and cat's-eye vortex formation in real time. The six-month historical record and unusual event log expose students to real observational data patterns across different magnetospheric regions.

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